individualism seem almost invincible. I carp but do congratulate this group of workers.

Perhaps two final quotes will be forgiven. The first is from Derrida's "Cogito and the History of Madness". It is as follows:-"The psychiatrist is but the delegate of this order, one delegate among others. Perhaps it does not suffice to imprison or to exile the delegate or to stifle him; and perhaps it does not suffice to deny oneself the conceptual material of psychiatry in order to exculpate one's own language. All our European languages of everything that has been participated, from far and near in the venture of Western reason- all this is the immense delegation of the project defined by Foucoult under the rubric of the capture and objectification of madness Nothing within this language, and no one among those who speak it, can escape the historical guilt - if there is one and if it is historical in a classical sense - which Foucault apparently wants to put on trial." The second quote is from the Portuguese poet Pessoa, and is as follows "In reality the unique critics of art and literature ought to be the psychiatrists; although they are as ignorant about these issues and remote from them, as what they call science as other people, nevertheless when faced with mental disease they have the competence that our judgement says they have. No body of human knowledge can be built on any other basis."

Dennett, D.C. (1996) Kinds of Minds: Towards an understanding of consciousness. Weidenfeld and Nicolson.

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Daniel Dennett's new book aims to introduce a general audience to current work in the evolutionary modelling of minds. This work, of which Dennett is one of the pioneers, is assuming increasing importance in contemporary cognitive science. A second, and equally important, aim of the book is to challenge the Cartesian, anthropomorphic and realist prejudices of the lay public.

Dennett begins, in his deceptively easy style, with a list of questions prompting the reader to consider which organisms might or might not possess minds. These questions are designed to bring readers up short and force them to think about where to draw the line:

'We left Houston at dawn, headin' down the road - just me and my truck.'
Strange. If this fellow thinks his truck is such a worthy companion that it deserves shelter under the umbrella of 'we' he must be very lonely. Either that, or his truck must have been customized in ways that would be the envy of roboticists everywhere. In contrast, 'we - just me and my dog' doesn't startle us at all, but 'we - just me and my oyster' is hard to take seriously. In other words, we're pretty sure that dogs have minds, and we're dubious that oysters do. (p. 4)

This problem is not only of theoretical interest, but also raises important ethical issues (pp. 4-5). In attributing mentality to a creature, we also grant it certain rights. Unfortunately, there is no agreement within the scientific and philosophical communities as to what the possession conditions for minds are.

Dennett begins by warning the reader away from what he takes to be a blind alley, and sketches once again his well-known, instrumentalist, view of the mind. To have a mind, he points out, is to have *intentional states* -- states with representational content. And it is easy to set the possession conditions for such states very high -- so high, according to some philosophers, that animals do not really count as having minds at all. This, Dennett thinks, is a mistake. Intentionality, he holds, is a widespread natural phenomenon, and human thought is just a fancier version of the sensitivities and tropisms exhibited by a simple systems such as plants, cells and primitive robots. We can think of such systems as pursuing goals, and registering environmental information that is relevant to attaining them. Of course, these systems do not explicitly *represent* the goals they seek; but the goals are nonetheless real, since they are the ones which evolutionary processes (or, in the case of robots, their human creators) have designed them to pursue. Such systems *have* reasons for their actions, but their reasons are, as Dennett puts it, 'free floating'. For example, a fledgling cuckoo ousts the eggs of its unwitting adoptive parents:

Why does it do this? Because those eggs contain rivals for the attentions of the surrogate providers. By disposing of these rivals, it maximizes the food and protective care it will receive. The newborn cuckoo, is of course oblivious, it has no inkling of this rationale for its ruthless act, but the rationale is there, and has undoubtedly shaped this innate behaviour over the eons... . (S)uch a rationale is "free floating" because it is nowhere represented in the fledging, or anywhere else... . The strategic principles involved are not explicitly encoded but just implicit in the larger organization of designed features. (p. 49)

Just as the cuckoo's "dumb", instinctive behaviour is a product of evolutionary design, so Dennett suggests, are our more sophisticated representational abilities.

Many philosophers draw a distinction between *derived* and *original* (or intrinsic) intentionality. The pictorial and written representations we produce have derived intentionality -- their content is derived from the intentions of their human users and producers. The content of our own thoughts, by contrasts, is not derived, but internally-generated. Dennett again demurs. All intentionality, he argues, external *and* internal, is derived:

... a mental image of your mother - or Michelle Pfeiffer - is about its object in just as derived a way as the sketch you draw. It is internal, not external, but it

is still an artefact created by your brain and means what it does because of a particular position in the ongoing economy of your brain's internal activities and their role in governing your body's complex activities in the real, surrounding world. (p. 52)

This immediately raises the question of just what role explicit internal representations (mental images of words and pictures) have in the mental economy. In order to shed some light on this question, Dennett adopts an evolutionary perspective.

He begins by introducing a metaphor: the evolution of the mind, he suggests, involved a progressive ascent of the *Tower of Generate and Test*. The Tower consists of four floors, each of which represents a more efficient way of solving day-to-day survival problems. Each progressive solution is a 'better move' than the one before. Thus, the ground floor is inhabited by *Darwinian* creatures that are blindly generated by natural selection and possess different hardwired phenotypes. Their responses to survival problems are determined by their genetic inheritance and are quite inflexible.

The second floor is inhabited by *Skinnerian* creatures. These can vary their phenotypic response to the environmental contingencies they encounter. Skinnerian creatures also possess hardwired reinforcement mechanisms that bias them to make what Dennett terms 'Smart Moves'. Such a creature will vary its response to stimuli until something good comes of it, whereupon it will become conditioned to produce the same response again should similar stimuli be encountered. Such conditioning is possible, of course, only if the initial response is not fatal.

Popperian creatures, who inhabit the third floor, run less risk of making fatal first moves. These creatures have an *inner environment* -- a mental representation of the external world -- and can run internal simulations of various courses of action. In this way, they can calculate the likely effect of candidate actions and eliminate the ones likely to have undesirable consequences -- thus 'permitting their hypotheses to die in their stead' as Karl Popper puts it.

Popperian creatures are much smarter than their Skinnerian cousins. However, their ability to form and test hypotheses is still limited by their genetic endowment. Their representational abilities, in particular, may remain relatively encapsulated, so that information from one domain is not routinely made available for the solution of problems occurring in others. *Gregorian* creatures, who live on the next floor, are smarter yet. They supplement their innate problem-solving abilities with *mind tools* acquired from their peers. They have learned Richard Gregory's lesson that tools not only *display* intelligence, but *create* it too. A well-designed tool meshes with our native abilities and extend them in new and far-reaching ways. (Think, for example, of how a pair of scissors extends our ability to manipulate and shape artefacts.) The *mind tools*, which Gregorian creatures possess are culturally transmitted tricks, shortcuts, and strategies which enable them to arrive more swiftly at Smart Moves for solving problems.

The most powerful of these mind tools, Dennett suggests, are *words*.

This picks up themes from Andy Clark's recent work. Clark argues that language augments existing computational abilities in several ways. For instance, language enables us to make plans and co-ordinate our actions. If you tell a friend you shall meet at 1 p.m. on Monday for lunch, then your friend can time her other activities accordingly. Language also permits the mental rehearsal of self-directed commands and exhortations in order to focus behaviour, aid recall, and reinforce learning. Clark writes:

The role of public language and text in human cognition is not limited to the preservation and communication of ideas. Instead, these external resources make available concepts, strategies and learning trajectories which are simply not available to individual, un-augmented brains. Much of the true power of language lies in its under appreciated capacity to reshape computational spaces which confront intelligent agents. (1995:18)

We are Gregorian creatures and the manipulation of words and other mind tools (*memes* as Dawkins calls them) is the distinctive mark of human mentality. But it is not a mark of mentality *as such*; mind tools are just another of Mother Nature's strategies for keeping us ahead in the complex game of survival, and they derive their content from their role in these strategies.

All of this is very attractive, and is well motivated by a desire to rid cognitive science of Cartesian and essentialist thinking. Dennett provides us with a good structure to pin our speculations upon, a way of avoiding the gritty problem of intrinsic (or original) intentionality, and some account of how the human mind differs from that of the not-so-flexible Popperian Creatures. All of this seems to provide a potentially fruitful frame for future empirical work.

But the lay-person, picking up this Science Masters publication, may be left with a nagging doubt. 'Granted', they will say, 'the reasons that motivate frogs and bats and cats are free-floating, not intrinsic; but surely *ours* aren't? We don't just act *for* reasons, we act *on* them.' This is a legitimate worry, and suggests that Dennett's model of derived intentionality is too loose as it stands. It is all very well to argue that our representational mechanisms have been selected for because they confer adaptive advantages, but the lay reader may wonder why these representation-generating mechanisms could not give us *intrinsic* intentionality also. And even if, as Dennett claims, intrinsic intentionality is illusory, it would be nice to have some explanation of why we dupe ourselves so thoroughly into believing in it.

The more expert reader, looking to see how the land lies within cognitive science, may end up asking the same question. Dennett's speculations on the role of natural language in the Gregorian mind may raise hopes of a more substantial solution to the problems of human intentionality. Might not our natural language

abilities be the origin of higher-order thinking and the source of real, cognitively potent, mental representations? Such hopes are dashed if language functions merely as a tool, facilitating and co-ordinating low-level tasks.

The later chapters of the book are less focused, but rewarding nonetheless. Here Dennett continues his exploration (begun in his *Consciousness Explained*) of the idea that the high-level structure of the human mind is determined by cultural programming, rather than biology. The human mind, he suggests, is highly organized community of low-level mechanisms in complex dynamical co-dependence with its environment. (He notes, for example, how some old people need their familiar home environment in order to function cognitively.) The final chapter is a vigorous and salutary polemic against woolly-minded anthropomorphism. The further reading section at the end is a wildly eclectic mix of stuff, some of which would baffle the general reader.

Those who have read Dennett's weightier tomes will learn little from this book, but anyone unfamiliar with his work will find it stimulating, provoking and illuminating. As usual, Dennett's avuncular tone makes for an easy read.

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There are 5 parts to this book. The first focuses mainly on the epidemiology of challenging behaviour. The second focuses on different recording systems used to analyse destructive behaviour. Parts 3 and 4 focus on intervention, including biological and environmental intervention. Finally, part 5 focuses on some ethical issues in treatments for destructive behaviour.

Different terminologies have been used to describe behaviour problems in people with learning difficulty. This terminology includes challenging behaviour, destructive behaviour, severe problem behaviour and extremely dangerous behaviour. The authors have not distinguished differences among them. One might wonder if a brief chapter on the definition of such terminologies would be valuable.

Nevertheless the lack of clear definitions does not cause any confusion in reading this book. It was written mainly by American colleagues and is thus composed of many American studies on mental retardation. However, they by no means ignore the contribution of studies from the United Kingdom. For instance, among the studies on the prevalence or epidemiology of destructive behaviour, Bruininks et al included research done by Felce et al, and Rojahn included research done by Oliver et al. Their incorporation of research from the United Kingdom makes the chapters become interesting to read because one can glimpse an international perspective on the investigation of challenging behaviour.

Since the publication of this book, few epidemiological studies carried out in the United Kingdom have been published (e.g. Qureshi, 1994). In other words, we are now in a good position to compare the recent epidemiological data in the United Kingdom with that in America. This book is undoubtedly a good reference for epidemiological studies on learning difficulties in America, perhaps a follow-up edition of this book should focus on the comparison between data in different countries, i.e. providing an international perspective on studying destructive behaviour with regard to the biological interventions for it.

While Nyhan has provided a good description of the clinical features of Lesch-Nyhan disease, he touched on the therapeutic aspects of this disease very briefly. He provided a table summarising different drug treatments on this disease but made no attempts to discuss any issues related to these drug trials. For this reason, this chapter gave the impression of being merely an introductory chapter on Lesch-Nyhan disease, rather than a discussion on biological intervention. This chapter would become a lot better if the author attempted to identify the strengths and weaknesses of the drug trials on this disease.

The last chapter on the ethical considerations in treatments of destructive behaviour is innovative. Sprague, after having briefly described his personal experience, introduced the ethics of human experimentation from a historical perspective, ranging from the Nazi's experiment, to the development of a code of conduct for research after World War II. He then talked about the nature, and the meaning, of efficacy, and treatment effectiveness. This chapter does raise some issues that are worthwhile considering as we are doing research on the evaluation of treatments. On the whole, this is a good reference book, and a good text book for destructive behaviour in developmental disabilities. Post-graduate students, clinicians, and researchers in developmental disabilities would find this book very useful indeed.

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